Aravind Karthik R

M +91 90925 33179 • **E** contactaravindkarthik@gmail.com • **in** aravindkarthikr
Chennai, India

Research Statement

My research is focused on a single question: *how do we detect when the brain's neurovascular and neuroelectrical systems are failing, and can we measure it early enough to matter clinically?* I pursue this through two complementary methodologies — functional biosignal acquisition and analysis (fNIRS, EEG) and quantitative computational imaging — applied to neurological and neurodevelopmental disorders including ASD, stroke, and Alzheimer's disease. My work spans the full stack from custom optical and electrophysiological front-ends to real-time embedded inference pipelines and rigorous statistical biomarker validation. My longer-term interest is in understanding how large-scale shifts in human behaviour — including AI tool adoption during development — alter functional brain organisation, measurable through the same fNIRS and EEG methods I have been building.

Education

Vellore Institute of Technology — **Chennai**
M.Tech, Embedded Systems — *Jul 2025 – Present*
Courses: Real-Time Operating Systems, Microcontroller Architecture, Machine Learning & Deep Learning, IoT Systems Design

Vellore Institute of Technology — **Chennai**
B.Tech, Electronics & Computer Engineering (Minor: Management) — *Jul 2021 – Sep 2025*
Courses: Data Structures & Algorithms, Embedded C, Artificial Intelligence, Machine Learning, Computer Networks, Embedded Systems

Manuscripts & Research Output

Mitochondrial Activity Scoring in Cells via Quantitative 3D-SIM Fluorescence Analysis — *In Preparation*
With **Prof. Ganesh Pandian Namasivayam** (iCeMS, **Kyoto University**, Japan). End-to-end morphometric pipeline; hold-out AUC = 0.891, Cohen's $d = 1.80$ ($p = 4.74 \times 10^{-9}$, Mann–Whitney U); 9/12 features at BH-corrected FDR 5%. Target: *Computers in Biology and Medicine / PLOS ONE.*

International Collaborations

Kyoto University — **Japan**
Research Collaboration — Quantitative Fluorescence Bioimaging — *2024 – Present*
With **Prof. Ganesh Pandian Namasivayam** (iCeMS). Quantitative morphometric analysis of mitochondrial network architecture in cancer cell biology using 3D-SIM super-resolution microscopy. Joint first-author manuscript in preparation.

Professional Experience

Indian Institute of Science (IISc) — **Bangalore**
Project Associate — BEES Lab, Supervisor: Prof. Hardik J. Pandya — *Jun 2024 – Jun 2025*
- Designed and deployed **real-time edge AI inference pipelines** on Raspberry Pi for biomedical computer vision; optimised model latency and memory footprint under hard embedded constraints.
- Architected end-to-end research workflows: raw NIR optical acquisition → dataset curation (>150 participants) → model training → quantisation → live deployment; achieved **12% improvement** in subcutaneous vein detection via DeepLabV3+ fine-tuning and presented processing demonstrations at **IISc Open Day 2025** to 500+ visitors across varied technical backgrounds.

Highbrow Diligence Services Pvt. Ltd. — **Chennai**
Embedded Systems Intern — *Aug 2023 – Dec 2023*
- Developed an embedded automation prototype for a medical **EBOO therapy device**; multi-sensor integration with safety-triggered actuator control via **SPI/I^{2}C** on Arduino-based platform.

Selected Research Projects

Mitochondrial Activity Score (MAS) Pipeline — M.Tech Thesis · Kyoto University Collaboration *Python · scikit-learn · scikit-image · OpenCV*

- End-to-end fluorescence analysis pipeline for **3D-SIM microscopy** (Zeiss AxioObserver, 63× oil, 0.03 μm/px, 2048×2048, 16-bit CZI) scoring mitochondrial network health in healthy vs. patient-derived metastatic cells; collaboration with **Kyoto University (iCeMS)**.
- Extracted 12 morphometric features (**MiNA framework**): junction-aware skeleton pruning, **Sauvola adaptive thresholding**, EDT core/edge decomposition, and Spearman-ρ Edge-Core Gradient Index.
- **L1-regularised logistic regression** in leakage-free `Pipeline`: hold-out AUC = 0.891, Cohen's $d = 1.80$ ($p = 4.74 \times 10^{-9}$, Mann–Whitney U); 9/12 features at BH-corrected FDR 5%.
- Quantified **DRP1-mediated fission**: 33% lower fragment count (946 vs. 1 410), 25% smaller network area (628 vs. 798 μm^2), 33% shorter skeleton (2 926 vs. 3 880 μm). *Methodological framework extensible to neuroimaging morphometry.*

fNIRS-Based Cerebral Haemodynamic Monitoring & Stroke Prediction *ESP32 · Raspberry Pi · TCP · TensorFlow Lite*

- Dual-wavelength optical front-end (780 nm / 820 nm IR LEDs with matched photodiodes) exploiting oxy/deoxy-Hb differential absorption for **non-invasive cerebral perfusion monitoring**; continuous haemodynamic signals reflect the integrity of **neurovascular coupling** — the physiological mechanism linking neural activity to blood flow.
- Haemodynamic signals streamed via TCP to Raspberry Pi for **edge ML** stroke-prediction inference; end-to-end sensing-to-prediction latency $\approx$250 ms; full on-device inference, zero cloud dependency.

NECTAR — RTL Design for Real-Time Neurovascular Coupling Anomaly Detection *Verilog · RTL Design · Icarus Verilog · GTKWave*

- Designed a custom **5-stage pipeline** in Verilog for real-time detection of neurovascular coupling anomalies — NVC breakdown is an early, cross-cutting signature in ASD, Alzheimer's disease, stroke, and traumatic brain injury, currently detectable only through offline, retrospective fNIRS analysis.
- Stage-wise RTL architecture: hardware noise filtering, parallel **DSP** preprocessing, low-latency feature extraction, hard-wired classification logic, and closed feedback loop; designed for <10 ms alarm-response latency.
- *Current status: behaviour-level RTL, functionally verified via Icarus Verilog/GTKWave. Physical implementation (synthesis, place-and-route, timing closure) targeting* ***Xilinx Artix-7*** *is the immediate next step; throughput and latency figures pending post-implementation validation.*

EEG-Based Computational Biomarker Analysis for Autism Spectrum Disorder *ESP32 · MCP3204 · Raspberry Pi · Signal Processing*

- Multi-channel EEG acquisition from 8 scalp sites (O1, O2, T3, T5, Fp1, Fp2, F4, F8) using ESP32 with MCP3204 12-bit ADC; analysis conducted against a **clinical EEG dataset from Sri Ramachandra Medical College & Research Institute (SRMC), Chennai** — ethically sourced, clinically diagnosed ASD and neurotypical cohorts.
- Extracted first-order (per-channel mean/std), second-order inter-channel covariance and cross-correlation matrices, and third-order skewness features; produced heatmaps, 3-D scatter plots, and distribution comparisons for quantitative **ASD–neurotypical group differentiation**.

Neurodevelopmental Speech Biomarker Analysis — Edge On-Device Pipeline *Raspberry Pi · FastDTW · MFCC · On-Device ML*

- Investigated prosodic and spectral speech features as computational biomarkers of neurodevelopmental disorder; **775 samples from speakers aged 8–14** — the primary diagnostic window for ASD and childhood language-processing conditions. Prosodic atypicality (pitch contour, rhythm, stress) is among the most consistently documented behavioural signatures in ASD.
- **44-dimensional feature vector** (spectral, MFCC, pitch, chroma, rhythm); **FastDTW** temporal alignment; 94% keyword recognition accuracy with full on-device inference and complete data privacy — zero cloud dependency in clinical-adjacent deployment.

Technical Skills

Programming: Embedded C, C++, Python, ARM Assembly (Thumb/IRQ), Verilog, MATLAB

Embedded & RTOS: **RTOS**, **QNX**, Linux Device Drivers, Embedded Linux, Firmware Dev, STM32, ESP32, Raspberry Pi, PSoC, NVIDIA Jetson, Arduino

AI & Edge Vision: TensorFlow, PyTorch, Scikit-learn, OpenCV, **TensorFlow Lite**, Model Quantisation, Edge Inference Optimisation, DeepLabV3+

Neuro & DSP: EEG/fNIRS Biosignal Processing, ADC Sampling, Digital Filtering, FFT, Sauvola Thresholding, Morphological Feature Extraction, **CUDA** Kernel Development

HW & Protocols: LoRa, BLE, SPI, I^{2}C, UART, MQTT, TCP/IP; PCB Schematic Validation, SMD Soldering

EDA & Tools: Quartus Prime, Icarus Verilog, GTKWave, Git, Jupyter, Matplotlib

Teaching, Outreach & Leadership

- **CUDA Seminar Lead,** VIT Chennai — Live GPU compute demonstration: kernel design, memory hierarchy, warp-level execution, and custom GPU driver authoring.
- **Programme Representative,** M.Tech Embedded Systems, VIT — Elected; coordinated scheduling across 6+ faculty; primary student–faculty liaison.
- **Research Demonstrator,** IISc Open Day 2025 — Live biosignal processing demonstrations to 500+ visitors.